

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination-2023

Course: B. Tech. Branch: Computer Engineering/CSE Semester: VII

Subject Code & Name: Artificial Intelligence (BTCOC701)

Max Marks: 60

Date: 02/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Elaborate Artificial intelligence with suitable example along with it's application.	Analyze	6
B) Discuss the historical evaluation of Artificial Intelligence.	Understand	6
C) State the relationship between Agent and Environment.	Remember	6
Q.2 Solve Any Two of the following.		12
A) Give the outline of Uniform cost search algorithm.	Apply	6
B) Explain A* algorithm for the shortest path.	Understand	6
C) What is alpha-beta pruning? Explain the function of alpha-beta pruning.	Understand	6
Q. 3 Solve Any Two of the following.		12
A) Explain the production based knowledge representation technique.	Understand	6
B) Give outline of simple forward Chaining algorithm with suitable example.	Remember	6
C) What is planning? What is need of planning.	Understand	6
Q.4 Solve Any Two of the following.		12
A) Compare and contrast the following: Forward Vs. Backward Reasoning	Remember	6
B) Explain Bayesian Network.	Understand	6
C) What is fuzzy set? Explain degree of membership. How does are represent a fuzzy set in a computer.	Apply	6

Q. 5	Solve Any Two of the following.		12
A)	What is Natural Language Processing? Explain various types of NLP techniques.	Understand	6
B)	What is Expert system? Explain its Architecture, Features & applications.	Understand	6
C)	Comparison between Syntactic Processing & Semantic Processing.	Remember	6

***** End *****